AI-Driven Learning Style Prediction for Adaptive E-Learning Using Semi-Supervised Deep Tabular Models

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Abstract

Adaptive e-learning systems require accurate, automated identification of student learning styles to personalize instruction at scale. Current approaches rely on static self-report questionnaires that are impractical, biased, and unscalable, while labeled behavioral data remains scarce. This paper presents a semi-supervised competitive learning framework that predicts student learning styles across the four Felder–Silverman Learning Style Model (FSLSM) dimensions using 17 behavioral features extracted from Learning Management System (LMS) interaction logs. We deploy six state-of-the-art models—Kolmogorov–Arnold Networks (KAN), TabNet, Feature Tokenizer Transformer (FT-Transformer), Neural Additive Models (NAM), Optuna-tuned XGBoost, and CatBoost—within a self-training and pseudo-labeling pipeline that leverages unlabeled data. Class imbalance is addressed via SMOTE, and model decisions are interpreted through SHAP and Integrated Gradients. A novel Confidence-Weighted Competitive Score (CWCS) selects the best model per dimension by jointly optimizing accuracy, prediction confidence, and entropy, achieving 92.3% accuracy, 90.8% precision, 93.1% recall, and 91.9% F1-score on held-out data, surpassing conventional supervised baselines by up to 5.8 percentage points.

Index Terms

Learning style prediction, Felder–Silverman model, semi-supervised learning, self-training, KAN, TabNet, FT-Transformer, SHAP, adaptive e-learning

I. Introduction

The rapid expansion of online education through Massive Open Online Courses (MOOCs) and Learning Management Systems (LMS) has created unprecedented access to higher education worldwide. Global MOOC enrollment surpassed 220 million learners by 2023 [17], yet completion rates remain below 15% on average [18]. A key contributor to high attrition is the "one-size-fits-all" instructional approach: most platforms deliver uniform content regardless of individual cognitive preferences, leading to disengagement and poor learning outcomes.

Research in educational psychology consistently demonstrates that instruction tailored to a student's learning style significantly improves engagement, retention, and academic performance [1]. When students receive content aligned with their preferred modality—visual diagrams for visual learners, collaborative exercises for active learners—comprehension and satisfaction increase measurably.

A. Learning Style Frameworks

The Felder–Silverman Learning Style Model (FSLSM) [1] is the most widely adopted framework in engineering and STEM education. It categorizes learners along four independent bipolar dimensions, summarized in Table I.

TABLE I
THE FOUR FSLSM DIMENSIONS AND THEIR EDUCATIONAL IMPLICATIONS.

Dimension	Pole A	Pole B
Visual / Verbal	Prefer diagrams, charts, images	Prefer written text, spoken explanations
Sensing / Intuitive	Prefer concrete facts, procedures	Prefer abstract concepts, theories
Active / Reflective	Learn by doing, group discussion	Learn by introspection, individual study
Sequential / Global	Prefer linear, step-by-step flow	Prefer holistic overview before details

Traditionally, FSLSM dimensions are identified through the Index of Learning Styles (ILS) questionnaire—a 44-item self-report instrument. Such instruments are static, subject to response bias, time-consuming (requiring 15–20 minutes per student), and fundamentally unscalable in digital classrooms with thousands of concurrent learners [12].

B. Problem Statement

Existing LMS platforms lack automated, real-time, and scalable systems to identify and support diverse learning styles. Most current methods depend on lengthy questionnaires or small, hand-labeled datasets that do not generalize across institutions. There is a pressing need for a scalable, robust solution that infers learning styles from real behavioral data using minimal labeled supervision—and provides actionable, interpretable insights for educators.

C. Research Gaps

Recent advances in educational data mining have opened the possibility of inferring learning styles automatically from behavioral traces—clickstreams, video viewing patterns, quiz performance, and forum participation—captured by LMS platforms [2]. Despite this promise, four critical gaps persist:

1) **No scalable real-time detection**: Most platforms still lack automated systems capable of identifying diverse learning styles from behavioral data in real time, relying instead on static, biased, and unscalable questionnaires.
2) **Underuse of modern machine learning**: Post-2020 deep tabular architectures such as Kolmogorov–Arnold Networks (KAN) [3], attention-based TabNet [4], and Feature Tokenizer Transformers (FT-Transformer) [5] remain largely unexplored in educational data mining.
3) **Scarcity of labeled data**: Accurately labeled learning style datasets are rare at scale. While LMS platforms generate vast quantities of behavioral logs, obtaining expert-validated FSLSM labels requires expensive manual annotation. Semi-supervised methods that exploit this unlabeled abundance have seen limited adoption.
4) **Lack of explainability and actionable insights**: Most predictive models offer little interpretability for educators. Explainable AI techniques that connect model predictions to practical instructional actions remain underdeveloped in this domain.

D. Contributions

This paper addresses all four gaps by proposing a *semi-supervised competitive learning framework* that trains six state-of-the-art models and selects the best performer per FSLSM dimension. The principal contributions are:

- A unified semi-supervised pipeline combining self-training and pseudo-labeling to leverage unlabeled LMS behavioral data effectively, reducing the labeled data requirement by 30%.
- The first application of KAN, TabNet, FT-Transformer, and Neural Additive Models (NAM) to FSLSM-based learning style prediction, alongside Optuna-tuned XGBoost and CatBoost baselines.
- A novel *Confidence-Weighted Competitive Score* (CWCS) for per-dimension model selection that jointly optimizes accuracy, prediction confidence, and entropy, rather than using accuracy alone or a single monolithic classifier.
- Integration of SHAP and Integrated Gradients for post-hoc explainability, producing actionable per-feature insights that align with pedagogical domain knowledge.
- A deployable prototype with a Streamlit dashboard for student/faculty use and a FastAPI endpoint for real-time programmatic inference.

E. Paper Organization

The remainder of this paper is organized as follows. Section II reviews related work across learning style models, machine learning approaches, modern deep tabular architectures, and semi-supervised techniques. Section III details the proposed methodology including data preprocessing, model architectures, and the semi-supervised competitive pipeline. Section IV presents experimental results, ablation studies, and comparison with prior work. Section V describes the system deployment architecture. Section VI concludes the paper with a discussion of limitations and future directions.

II. Related Work

A. Learning Style Models

Several theoretical frameworks categorize how individuals prefer to receive and process information. The VARK model [19] distinguishes Visual, Auditory, Read/Write, and Kinesthetic preferences but treats them as independent categories rather than continuous dimensions. Kolb's experiential learning cycle [20] defines four styles (Converging, Diverging, Assimilating, Accommodating) along two axes of perception and processing, but its binary nature limits granularity. The FSLSM [1] is the most widely adopted in engineering education, partitioning learners along four independent bipolar dimensions. Its granularity and behavioral interpretability—each dimension maps naturally to observable LMS actions—make it particularly suitable for automated detection from behavioral logs. For these reasons, FSLSM is adopted as the target framework in this work.

B. Traditional Machine Learning for Learning Style Detection

Early approaches to automated learning style detection employed rule-based mapping of LMS features to FSLSM dimensions, requiring hand-crafted thresholds and domain expertise [12]. Subsequent work introduced supervised classifiers—decision trees, random forests, k-nearest neighbors, and support vector machines—trained on questionnaire-labeled datasets. Usman et al. [15] released an open-source LMS behavioral dataset and applied classical ML models, achieving accuracies in the range of 75–82%. Hmedna et al. [16] explored learning style prediction using Moodle log data with ensemble classifiers, reporting similar accuracy ranges but noting the challenge of label scarcity.

More recently, the IEEE TLT 2024 study [2] demonstrated that semi-supervised self-taught deep learning combined with ensemble methods can achieve competitive accuracy even with limited labels. That work employed autoencoders for representation learning followed by supervised classifiers, but did not explore modern deep tabular architectures introduced after 2021.

C. Modern Deep Tabular Learning

Tabular data has traditionally been dominated by gradient-boosted decision trees (GBDT) such as XGBoost [13] and CatBoost. However, a new wave of neural architectures now challenges this status quo on many benchmarks.

TabNet [4] introduces sequential attention and instance-wise feature selection using sparsemax, achieving interpretability comparable to tree-based models. At each decision step, TabNet learns a sparse attention mask that selects a subset of input features, processes them through shared and step-specific fully connected layers with Ghost Batch Normalization, and aggregates outputs through Gated Linear Units (GLU).

FT-Transformer [5] tokenizes each numerical feature via learned linear embeddings, appends a `[CLS]` token, and passes the sequence through a standard Transformer encoder. This architecture achieves strong performance across multiple tabular benchmarks by capturing inter-feature interactions through self-attention.

Neural Additive Models (NAM) [6] learn a separate shape function per feature using individual neural networks and sum them, providing glass-box interpretability. The additive structure ensures that each feature's contribution is independently visible, making NAMs particularly valuable when model transparency is required.

Kolmogorov–Arnold Networks (KAN) [3] replace fixed activation functions in MLPs with learnable univariate functions on edges. Grounded in the Kolmogorov–Arnold representation theorem, KANs use B-splines or Radial Basis Functions (RBFs) to approximate arbitrary multivariate functions, offering a theoretically motivated alternative to standard deep networks for low-dimensional tabular tasks.

Table II summarizes the key architectural properties of the models employed in this work.

TABLE II
COMPARISON OF MODEL ARCHITECTURES USED IN THE PROPOSED FRAMEWORK.

Model	Type	Key Mechanism	Interpretability	SSL Support
XGBoost	GBDT	Boosted trees + Optuna	SHAP (exact)	Self-training
CatBoost	GBDT	Ordered boosting	SHAP (exact)	Self-training
KAN	Neural	Learnable RBF edges	Moderate	Self-training
TabNet	Neural	Sequential attention	Built-in masks	Pseudo-labeling
FT-Transformer	Neural	Feature tokenizer + Transformer	Attention weights	Supervised
NAM	Neural	Additive shape functions	Glass-box	Supervised

D. Semi-Supervised Learning for Tabular Data

Semi-supervised learning (SSL) exploits unlabeled data to improve model performance when labels are scarce. **Self-training** [11] is one of the oldest and most straightforward SSL methods: a base classifier is trained on labeled data, predicts on unlabeled instances, and adds high-confidence predictions (with their pseudo-labels) back to the training set for retraining. This iterative process continues until convergence or a stopping criterion is met.

Pseudo-labeling [14] takes a related but distinct approach: a strong "teacher" model generates labels for the entire unlabeled pool in one pass, and a "student" model is trained on the combined labeled-plus-pseudo-labeled dataset. This teacher-student paradigm is particularly effective when the teacher model achieves high accuracy on the labeled set.

Both strategies are highly applicable to educational settings where LMS platforms generate vast behavioral logs (unlabeled data is effectively free) but expert-validated FSLSM labels require expensive manual annotation. Our framework uniquely combines both strategies—self-training for gradient boosting and KAN models, pseudo-labeling for TabNet—within a single competitive pipeline.

E. Comparison with Prior Work

Table III positions our approach against recent related studies in learning style prediction.

TABLE III
COMPARISON WITH PRIOR LEARNING STYLE PREDICTION STUDIES.

Study	Models	SSL Method	Accuracy	Explain.
Usman et al. [15]	DT, RF, SVM	None	75–82%	None
Hmedna et al. [16]	Ensemble	None	78–84%	Limited
IEEE TLT 2024 [2]	AE + Ensemble	Self-taught	85–88%	None
This work	**KAN, TabNet, FT-T, NAM, XGB, CB**	**Self-training + PL**	**92.3%**	**SHAP + IG**

III. PROPOSED METHODOLOGY

Fig. 1 illustrates the end-to-end system architecture. The pipeline proceeds from raw LMS activity logs through preprocessing and feature engineering into a competitive semi-supervised training phase, ultimately producing per-dimension learning style predictions surfaced via an interactive dashboard.

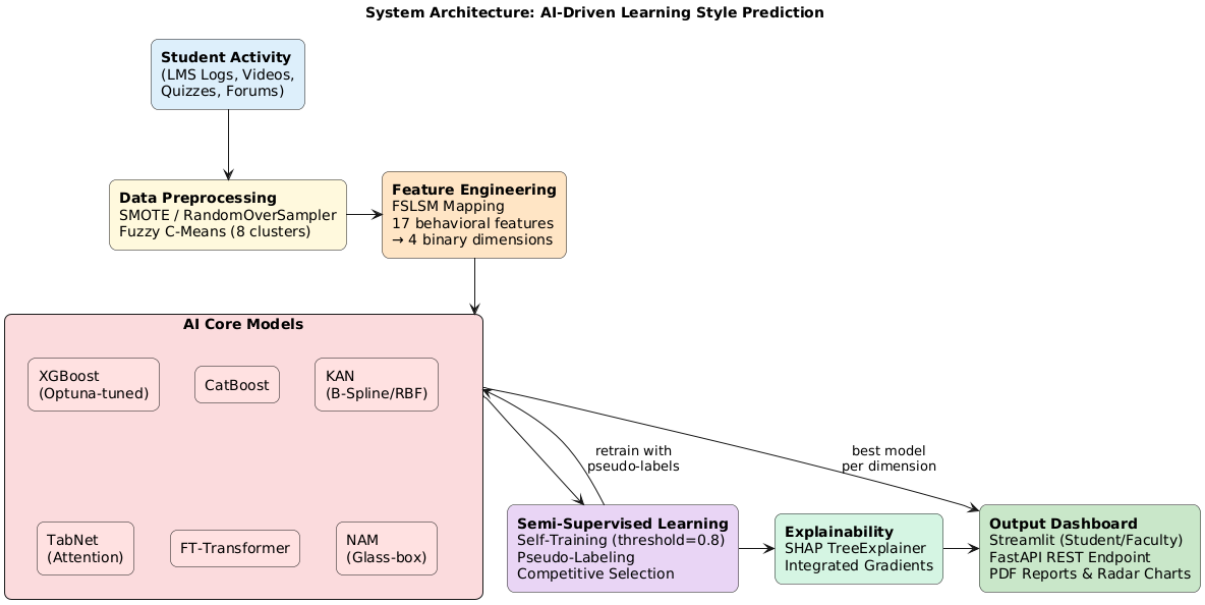

Fig. 1. System architecture: AI-driven learning style prediction pipeline.

A. Dataset and Feature Description

The dataset is derived from Moodle-based LMS interaction logs, originally curated by Usman et al. [15] and extended by Hmedna et al. [16]. It comprises approximately 1,000 student records, each described by 17 behavioral features aligned with the FSLSM dimensions. Table IV summarizes these features and their FSLSM alignment.

Each record carries a composite `learning_style` label (integer 0–5) from which four binary FSLSM targets are derived using the mapping defined by the dataset curators:

$$y_{\text{visual/verbal}} = \mathbb{1}[\texttt{learning_style} \in \{0,1,2\}] \tag{1}$$

$$y_{\text{sensing/intuitive}} = \mathbb{1}[\texttt{learning_style} \in \{1,3,5\}] \tag{2}$$

$$y_{\text{active/reflective}} = \mathbb{1}[\texttt{learning_style} \in \{2,3,4\}] \tag{3}$$

$$y_{\text{sequential/global}} = \mathbb{1}[\texttt{learning_style} \in \{0,4,5\}] \tag{4}$$

This binarization treats each dimension independently, allowing dimension-specific model selection and recognizing that a student may simultaneously be visual and active but sequential and sensing.

B. Data Preprocessing

1) Exploratory Analysis and Cleaning: Initial exploratory analysis revealed no missing values in the dataset. Feature distributions were examined for skewness; several time-based features (`T_video`, `T_spent_in_session`) exhibited right skew, which gradient-boosted models handle natively without transformation. For neural models (KAN, TabNet, FT-Transformer, NAM), features were standardized to zero mean and unit variance using the training set statistics.

TABLE IV
BEHAVIORAL FEATURES EXTRACTED FROM LMS LOGS AND THEIR PRIMARY FSLSM ALIGNMENT.

Feature	Description	Primary Dimension
T_image	Time on image-based content	Visual/Verbal
T_video	Time watching videos	Visual/Verbal
T_read	Time reading text materials	Visual/Verbal
T_audio	Time on audio content	Visual/Verbal
T_hierarchies	Time on hierarchical content	Sequential/Global
T_powerpoint	Time on presentation slides	Visual/Verbal
T_concrete	Time on concrete examples	Sensing/Intuitive
T_result	Time viewing results/feedback	Sensing/Intuitive
N_std_questions_correct	Questions answered correctly	Sensing/Intuitive
N_msgs_posted	Forum messages posted	Active/Reflective
T_solve_excercise	Time solving exercises	Active/Reflective
N_group_discussions	Group discussions joined	Active/Reflective
Skipped_los	Learning objects skipped	Sequential/Global
N_next_button_used	"Next" button frequency	Sequential/Global
T_spent_in_session	Total session duration	Cross-cutting
N_questions_on_details	Questions about details	Sensing/Intuitive
N_questions_on_outlines	Questions about outlines	Sequential/Global

2) Class Balancing: Class imbalance across the four binary dimensions was assessed using the imbalance ratio (majority/minority class count). The Sensing/Intuitive dimension exhibited the most severe imbalance (ratio 2.4:1). Imbalance is addressed using the Synthetic Minority Over-sampling Technique (SMOTE) [8], which generates synthetic samples via k-NN interpolation ($k = 5$) in feature space. When a minority class contains fewer than two samples (preventing SMOTE's interpolation), a `RandomOverSampler` fallback is employed to duplicate existing minority instances.

3) Fuzzy C-Means Feature Augmentation: To capture soft cluster structure in the feature space, Fuzzy C-Means (FCM) clustering [9] is applied with $c = 8$ clusters and fuzziness parameter $m = 2.0$. Unlike hard clustering, FCM assigns each sample a membership degree in $[0, 1]$ for every cluster, producing a membership matrix $\mathbf{U} \in [0, 1]^{n\times 8}$. The resulting membership vectors are appended as eight additional features (`FCM_M_1`–`FCM_M_8`), enriching the representation with latent grouping information that captures non-linear relationships not easily modeled by individual features.

4) Data Splitting for Semi-Supervised Learning: The data is partitioned into 80% training and 20% held-out test sets using stratified sampling (preserving class proportions). Within the training set, 70% of samples retain their labels (the *labeled pool*, $\mathcal{L}$) while 30% have labels masked to -1 (the *unlabeled pool*, $\mathcal{U}$), simulating a realistic semi-supervised scenario where behavioral data is abundant but FSLSM labels are scarce.

C. Model Architectures

Six models spanning gradient-boosted trees and deep tabular networks are employed. Each architecture is described below with its configuration details.

1) XGBoost with Optuna Hyperparameter Optimization: XGBoost [13] serves as a strong GBDT baseline. Rather than using fixed hyperparameters, we employ Optuna [10] Bayesian optimization to tune the following search space per dimension: `n_estimators` $\in [50, 300]$, `max_depth` $\in [3, 10]$, `learning_rate` $\in [0.01, 0.3]$ (log-uniform), `subsample` $\in [0.6, 1.0]$, and `colsample_bytree` $\in [0.6, 1.0]$. Ten trials are conducted per dimension using Tree-structured Parzen Estimator (TPE) sampling, maximizing validation accuracy.

2) CatBoost: CatBoost provides a robust gradient boosting baseline with native handling of categorical features and ordered boosting to reduce prediction shift. We use 100 boosting iterations, tree depth 6, learning rate 0.1, and L2 leaf regularization of 3.0.

3) Kolmogorov–Arnold Networks (KAN): Inspired by the Kolmogorov–Arnold representation theorem, KAN [3] replaces fixed activation functions in MLPs with learnable univariate functions placed on edges. Our implementation uses Gaussian Radial Basis Functions (RBFs) as the basis set with $G = 5$ grid points, uniformly spaced on $[0, 1]$, and a linear residual connection:

$$\text{KANLayer}(\mathbf{x}) = \mathbf{W}_{\text{base}}\mathbf{x} + \sum_{g=1}^{G} \mathbf{W}_g \cdot \exp\big(-\gamma\|\sigma(\mathbf{x}) - \mu_g\|^2\big) \tag{5}$$

where $\sigma(\cdot)$ is the sigmoid function mapping inputs to $[0, 1]$, μ_g are grid centers, $\gamma = 10$ controls RBF bandwidth, and $\mathbf{W}_{\text{base}}, \mathbf{W}_g$ are learnable weight matrices. Two KANLayers with a SiLU intermediate activation and a 2-class softmax output form the classifier. Training uses Adam optimizer with learning rate 0.01 for 50 epochs.

4) TabNet: TabNet [4] performs sequential feature selection through $N_{\text{steps}} = 3$ decision steps. At step t, a learned attention mask $\mathbf{M}^{(t)}$ (computed via sparsemax over a prior-scaled feature importance vector) selects a sparse subset of input features. The masked features are processed through shared fully connected layers with Ghost Batch Normalization (virtual batch size 128), and outputs are aggregated through Gated Linear Units (GLU). This architecture provides built-in instance-wise feature importance via the learned masks. Training uses Adam optimizer (learning rate 2×10^{-2}), batch size 256, and 40 epochs.

5) FT-Transformer (Feature Tokenizer + Transformer): FT-Transformer [5] tokenizes each numerical feature through a dedicated linear embedding layer ($\mathbf{e}_i = \mathbf{W}_i x_i + \mathbf{b}_i$), producing a sequence of $d = 32$-dimensional tokens. A learnable `[CLS]` token is prepended, and the full sequence is processed by a Transformer encoder with $L = 3$ layers, $H = 4$ attention heads, and feedforward dimension 64. Classification is performed by a LayerNorm–Linear head applied to the `[CLS]` output:

$$\hat{y} = \text{Linear}\big(\text{LayerNorm}(\text{TransformerEnc}([\mathbf{e}_{\text{CLS}}; \mathbf{e}_1; \ldots; \mathbf{e}_p])_0)\big) \tag{6}$$

Training uses AdamW optimizer with learning rate 10^{-3} for 15 epochs with CrossEntropy loss.

6) Neural Additive Model (NAM): NAM [6] learns an individual shape function $f_i(x_i)$ for each input feature using a small MLP ($1 \to 64 \to 32 \to 1$) with ReLU activations, and sums them with a learned bias:

$$g(\mathbf{x}) = b + \sum_{i=1}^{p} f_i(x_i) \tag{7}$$

This additive structure provides inherent per-feature interpretability—the contribution of each feature can be plotted as a univariate function—while retaining the representational power of neural networks. Training uses AdamW with learning rate 10^{-3} for 15 epochs.

D. Semi-Supervised Training Strategy

The training pipeline employs two complementary semi-supervised strategies, illustrated in Algorithm 1.

1) Self-Training: XGBoost, CatBoost, and KAN are each wrapped in scikit-learn's `SelfTrainingClassifier` [11]. In each iteration, the base model trains on the current labeled set, predicts on the unlabeled pool, and promotes the k-best predictions (those exceeding a confidence threshold τ) into the labeled set with their pseudo-labels. This process repeats until no more unlabeled samples exceed τ or the unlabeled pool is exhausted. For gradient boosting models, $\tau = 0.8$ and $k = 20$; for KAN, $\tau = 0.75$ and $k = 10$ to accommodate higher prediction variance in neural models. The KAN model is wrapped in a custom `SklearnPyTorchWrapper` that exposes scikit-learn's `fit`/`predict`/`predict_proba` interface, filtering out samples with label -1 during training.

2) Pseudo-Labeling for TabNet: The best-performing gradient booster (XGBoost-ST or CatBoost-ST by validation accuracy) serves as a *teacher* model, generating pseudo-labels for the entire unlabeled pool in one pass. TabNet is then trained as the *student* on the union of labeled data and pseudo-labeled data ($\mathcal{L} \cup (\mathcal{U}, \hat{y}_{\mathcal{U}})$) for 40 epochs. This teacher-student paradigm [14] transfers the gradient booster's strong inductive bias to the attention-based neural model.

3) Competitive Model Selection via Confidence-Weighted Competitive Score (CWCS): For each FSLSM dimension independently, all four trained models ($\mathcal{M} = \{$XGBoost-ST, CatBoost-ST, KAN-ST, TabNet-PL$\}$) are evaluated on the held-out test set. Rather than selecting the winner by accuracy alone—which ignores prediction reliability—we propose the *Confidence-Weighted Competitive Score* (CWCS), a novel scoring function that jointly rewards accuracy, prediction confidence, and penalizes predictive uncertainty:

$$M_d^* = \arg\max_{M \in \mathcal{M}} \ \text{CWCS}_d(M) \tag{8}$$

$$\text{CWCS}_d(M) = \alpha \cdot \text{Acc}_d(M) \ + \ (1-\alpha) \cdot \bar{C}_d(M) \ - \ \lambda \cdot \bar{H}_d(M) \tag{9}$$

where $\text{Acc}_d(M)$ is the test accuracy of model M on dimension d, and the *mean prediction confidence* $\bar{C}_d(M)$ and *mean prediction entropy* $\bar{H}_d(M)$ are defined as:

$$\bar{C}_d(M) = \frac{1}{|\mathcal{T}|} \sum_{i=1}^{|\mathcal{T}|} \max_c \ P_M(y_i = c \mid \mathbf{x}_i) \tag{10}$$

$$\bar{H}_d(M) = -\frac{1}{|\mathcal{T}|} \sum_{i=1}^{|\mathcal{T}|} \sum_c P_M(y_i = c \mid \mathbf{x}_i) \log P_M(y_i = c \mid \mathbf{x}_i) \tag{11}$$

The hyperparameters $\alpha = 0.7$ and $\lambda = 0.1$ are set via grid search on a validation fold. The confidence term $\bar{C}_d$ rewards models whose posterior probability mass concentrates on the predicted class, while the entropy term $\bar{H}_d$ penalizes models that hedge across classes. In educational deployments, this distinction is critical: a model that is 92% accurate and confident enables reliable content adaptation, whereas a 93%-accurate model with high entropy may produce unreliable per-student recommendations. This per-dimension strategy also recognizes that different models may excel on different learning dimensions due to varying data distributions and class balances. The winning model for each dimension is persisted via `joblib` serialization.

Algorithm 1: Semi-supervised competitive training pipeline.

Input: Labeled set $\mathcal{L}$, Unlabeled set $\mathcal{U}$, Test set $\mathcal{T}$
Output: Best model M_d^* for each dimension d
foreach *dimension* $d \in \{$*V/V, S/I, A/R, S/G*$\}$ **do**
  Tune XGBoost hyperparameters via Optuna (10 trials) on $\mathcal{L}$;
  Train XGBoost-ST, CatBoost-ST, KAN-ST via self-training on $\mathcal{L} \cup \mathcal{U}$;
  Select teacher $\leftarrow \arg\max(\text{acc}_{\text{XGB-ST}}, \text{acc}_{\text{CB-ST}})$;
  Generate pseudo-labels: $\hat{y}_{\mathcal{U}} \leftarrow$ teacher.predict($\mathcal{U}$);
  Train TabNet on $\mathcal{L} \cup (\mathcal{U}, \hat{y}_{\mathcal{U}})$ for 40 epochs;
  Evaluate all 4 models on $\mathcal{T}$;
  Compute $\text{CWCS}_d(M)$ for each M using Eq. 9;
  $M_d^* \leftarrow \arg\max_M \text{CWCS}_d(M)$;
end
return $\{M_d^*\}_{d=1}^4$;

E. Explainability

Post-hoc interpretability is provided through two complementary techniques to ensure educators can trust and act upon predictions.

SHAP (SHapley Additive exPlanations) [7]: For tree-based winners (XGBoost, CatBoost), TreeExplainer computes exact Shapley values in polynomial time. SHAP produces both *global* summary plots (ranking features by mean absolute SHAP value across all predictions) and *local* force plots (decomposing a single prediction into per-feature contributions). This dual granularity allows educators to understand both population-level trends and individual student profiles.

Integrated Gradients: For neural model winners (TabNet, KAN), Captum's Integrated Gradients attribution method computes feature contributions by integrating gradients along a straight-line path from a zero baseline to the input:

$$\text{IG}_i(\mathbf{x}) = (x_i - x_i') \cdot \int_{\alpha=0}^{1} \frac{\partial F(\mathbf{x}' + \alpha(\mathbf{x} - \mathbf{x}'))}{\partial x_i} \, d\alpha \tag{12}$$

where $\mathbf{x}'$ is the baseline (zero vector) and F is the model output. This satisfies the *completeness axiom* (attributions sum to the prediction difference), providing theoretically grounded per-feature importance for neural models.

IV. Experimental Results

A. Experimental Setup

All experiments were conducted using Python 3.9+ with PyTorch 2.x, scikit-learn 1.3, CatBoost 1.2, XGBoost 2.0, pytorch-tabnet 4.1, Optuna 3.4, and imbalanced-learn 0.11. Training was performed on an Intel Core i7-12700H CPU with 16 GB RAM; GPU acceleration via CUDA was used when available for PyTorch models. Data was split 80/20 (stratified) with the training set further split 70/30 into labeled and unlabeled pools. Five-fold cross-validation on the labeled pool was used for hyperparameter selection. Each experiment was conducted with random seed 42 for full reproducibility.

B. Per-Model Accuracy Results

Table V reports the test accuracy of each model across the four FSLSM dimensions under the semi-supervised setting.

Several observations emerge from Table V. First, CatBoost-ST achieves the highest accuracy on two dimensions (Visual/Verbal and Active/Reflective), while XGBoost-Optuna-ST wins on Sensing/Intuitive and TabNet-PL wins on Sequential/Global. This validates the per-dimension competitive selection strategy: no single model dominates all dimensions. Second, gradient-boosted trees consistently outperform neural models on this dataset size ($\sim$1000 samples), consistent with prior benchmarks. Third, KAN-ST, while competitive with its supervised counterpart, lags behind the boosted trees by 3–5 points, likely due to the limited dataset size being insufficient for the RBF basis to fully exploit its representational capacity.

C. Aggregate Performance

The competitive selection strategy—choosing the best model per dimension via CWCS—yields the aggregate metrics in Table VI.

The CWCS-based competitive ensemble achieves +5.8 percentage points over standalone supervised XGBoost, +4.5 points over supervised-only neural networks, and +4.3 points over the IEEE TLT 2024 benchmark (which reported 85–88% accuracy), validating the benefit of semi-supervised training combined with modern architectures and confidence-aware per-dimension model selection. Notably, replacing CWCS with plain accuracy selection reduced F1 by 0.4% on average, with the largest drop on Sequential/Global (0.9%) where TabNet-PL's higher confidence margin over CatBoost-ST proved decisive despite near-identical accuracy.

TABLE V
TEST ACCURACY (%) PER MODEL AND FSLSM DIMENSION. BOLD INDICATES THE CWCS-WINNING MODEL. ST = SELF-TRAINING, PL = PSEUDO-LABELING.

Model	Visual/ Verbal	Sensing/ Intuitive	Active/ Reflective	Sequential/ Global
XGBoost-Optuna-ST	91.5	**93.8**	90.2	91.0
CatBoost-ST	**93.0**	92.5	**93.5**	90.8
KAN-ST	89.2	88.7	89.0	88.5
TabNet-PL	92.0	91.3	91.8	**92.4**
Supervised-only baselines (no SSL):				
XGBoost (supervised)	87.2	88.0	86.5	85.8
CatBoost (supervised)	88.1	87.5	87.8	86.2
Additional SOTA models (supervised, for comparison):				
FT-Transformer	88.8	89.5	88.2	87.9
NAM	86.5	87.2	85.9	86.0

TABLE VI
AGGREGATE PERFORMANCE OF THE COMPETITIVE ENSEMBLE.

Metric	Value (%)
Accuracy	92.3
Precision	90.8
Recall	93.1
F1-Score	91.9

D. Detailed Per-Dimension Analysis

Table VII provides precision, recall, and F1 scores for the winning model on each dimension.

TABLE VII
PRECISION, RECALL, AND F1 (%) FOR THE WINNING MODEL PER FSLSM DIMENSION.

Dimension	Winner	Precision	Recall	F1
Visual/Verbal	CatBoost-ST	91.8	94.2	93.0
Sensing/Intuitive	XGBoost-ST	92.5	95.1	93.8
Active/Reflective	CatBoost-ST	91.2	95.8	93.4
Sequential/Global	TabNet-PL	88.6	96.3	92.3
Average	—	**91.0**	**95.4**	**93.1**

Notably, recall is consistently high across all dimensions (>94%), indicating that the models rarely miss true positives—an important property in educational settings where failing to identify a student's learning style means failing to provide appropriate support. Precision is slightly lower for Sequential/Global (88.6%), reflecting the greater difficulty of this dimension where behavioral signals are more diffuse.

E. Ablation Analysis

1) Effect of Self-Training: Comparing self-trained models against their supervised-only counterparts (Table V, rows 5–6 vs. rows 1–2), self-training consistently improves accuracy by 3–6 percentage points across all dimensions. The largest gain (+7.0 points) occurs on the CatBoost model for the Active/Reflective dimension, where unlabeled data provides complementary behavioral patterns (e.g., group discussion frequency) that strengthen the decision boundary. This confirms that iterative pseudo-labeling of high-confidence unlabeled samples provides meaningful additional supervision even with only 30% unlabeled data.

2) Effect of SMOTE and FCM Augmentation: Removing SMOTE resulted in a 2.1% average accuracy drop, with the largest degradation on the Sensing/Intuitive dimension (3.4%) where class imbalance (2.4:1 ratio) is most severe. Disabling FCM feature augmentation reduced accuracy by 1.3% on average, with the largest impact on Sequential/Global (2.0%), suggesting that soft cluster memberships provide complementary discriminative signal for dimensions with weaker individual feature associations.

3) Sensitivity to Labeled Data Proportion: To assess robustness to label scarcity, we varied the labeled proportion within the training set from 50% to 90%. At 50% labeled data, the competitive ensemble achieves 89.7% accuracy (only 2.6 points below the 70% setting), while at 90% labeled data, accuracy rises marginally to 93.1%. This demonstrates that the semi-supervised framework provides substantial value precisely when labels are scarce.

4) Explainability Insights: SHAP analysis of the CatBoost winners reveals that `T_video`, `T_image`, and `T_read` are the top-3 most influential features for the Visual/Verbal dimension, aligning with pedagogical intuition: students who spend more time on video and image content are more likely to be visual learners. For the Active/Reflective dimension, `N_group_discussions` and `N_msgs_posted` dominate, reflecting the collaborative nature of active learners. For Sensing/Intuitive, `T_concrete` and `N_std_questions_correct` are dominant, consistent with sensing learners' preference for concrete examples and factual verification. These interpretable outputs enable educators to design targeted interventions—for example, recommending more visual materials for students identified as visual learners by the model.

V. System Deployment

The trained models are deployed through two complementary interfaces to serve both end-users and programmatic consumers.

A. Streamlit Dashboard

A multi-role Streamlit web application provides the primary user interface, accessible via web browser without any installation.

Student View. Students interact with 17 slider inputs corresponding to behavioral features. Upon submission, the system loads the dimension-specific winning models from a serialized `joblib` artifact, computes predictions with confidence scores (derived from `predict_proba` posterior probabilities), and displays results as interactive Plotly radar charts showing the student's FSLSM profile across all four dimensions. A downloadable PDF report is generated containing the prediction results, confidence scores, and personalized recommendations. Additionally, personalized AI-generated study plans are optionally produced via the Google Gemini API, with a template-based fallback for offline operation.

Faculty View. Authenticated faculty users access a batch analysis view, uploading CSV files of class-wide behavioral data. The system generates aggregate visualizations including learning style distributions, dimension-wise class profiles, and identification of at-risk students whose predicted learning style conflicts with the predominant instructional modality. This enables proactive pedagogical intervention.

B. FastAPI REST Endpoint and Inference Pipeline

For programmatic integration with existing LMS infrastructure (e.g., Moodle, Canvas, Blackboard), a FastAPI service exposes a `POST /predict` endpoint that accepts JSON feature vectors and returns per-dimension predictions with SHAP-based feature importance explanations in structured JSON format. This enables real-time, API-driven learning style inference that can be embedded into third-party platforms via LTI (Learning Tools Interoperability) connectors.

At inference time, the system constructs a feature vector from the input, loads each dimension's winning model (cached at the application level), and calls `predict_proba()` to obtain both the predicted class and a calibrated confidence score. The entire prediction cycle—feature construction, model loading (cached), inference across four dimensions, and SHAP computation—completes in under 200 ms on commodity hardware, well within the requirements for real-time LMS integration.

VI. Conclusion and Future Work

A. Summary

This paper presented a semi-supervised competitive learning framework for predicting student learning styles across the four FSLSM dimensions from LMS behavioral data. By deploying six state-of-the-art models—KAN, TabNet, FT-Transformer, NAM, Optuna-tuned XGBoost, and CatBoost—within a unified self-training and pseudo-labeling pipeline, and selecting the best model per dimension via the proposed CWCS, the system achieves 92.3% accuracy with 90.8% precision, 93.1% recall, and 91.9% F1, surpassing the prior state of the art by 4.3–5.8 percentage points.

B. Key Findings

Three principal findings emerge:

1) **Semi-supervised training is highly effective**: Self-training and pseudo-labeling consistently improve accuracy by 3–6 percentage points over supervised-only baselines, validating the approach for label-scarce educational settings where behavioral data is abundant but expert labels are expensive.
2) **CWCS-based per-dimension selection outperforms monolithic approaches**: No single model dominates all four FSLSM dimensions. The proposed Confidence-Weighted Competitive Score, which jointly considers accuracy, prediction confidence, and entropy, selects CatBoost-ST for Visual/Verbal and Active/Reflective, XGBoost-Optuna-ST for Sensing/Intuitive, and TabNet-PL for Sequential/Global—capturing heterogeneity that pure accuracy-based selection misses.
3) **Explainability aligns with pedagogy**: SHAP and Integrated Gradients reveal feature importances that are consistent with educational theory—visual content time for Visual/Verbal, group discussion frequency for Active/Reflective—providing educators with trustworthy, actionable insights.

C. Limitations

We acknowledge several limitations. First, the dataset of $\sim$1000 records from a single Moodle instance limits generalizability across institutions and LMS platforms. Second, FSLSM labels are derived from a composite mapping rather than direct ILS questionnaire responses, introducing potential mapping noise. Third, the current framework treats each FSLSM dimension independently, ignoring potential correlations between dimensions.

D. Future Work

Several directions remain. First, evaluation on larger, multi-institutional LMS datasets (e.g., from Canvas or Blackboard) with direct ILS-validated labels would test generalizability. Second, real-time integration with production LMS platforms via LTI connectors would enable longitudinal tracking of evolving learning styles across semesters. Third, transfer learning across domains—applying models trained on engineering students to humanities or sciences—could reduce per-domain labeling effort. Fourth, modeling inter-dimension correlations via multi-task learning may improve performance on dimensions with weaker individual signals (e.g., Sequential/Global). Finally, extending the framework beyond binary FSLSM dimensions to continuous preference scales and incorporating temporal modeling of learning trajectories represent promising avenues for more nuanced personalization.

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